Gated Score-Residual Adaptation for Efficient Cross-Domain Named Entity Recognition

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ABSTRACT

Cross-domain named entity recognition (NER) must learn specialist entity inventories from limited target-domain supervision without discarding transferable span evidence. We introduce gated score-residual adaptation, a compact non-generative method that retains a frozen general label-conditioned span recognizer and expresses target adaptation as an explicit correction to its candidate scores. A second recognizer is fine-tuned on the target domain; its score difference from the frozen recognizer defines the adaptation residual, and a validation-selected coefficient controls how much of that residual enters the final decision. The construction separates reusable evidence from domain correction and requires no additional neural parameters after adaptation. Experiments cover all five specialist domains of CrossNER and use strict typed-span micro-precision, recall, and F1. Domain adaptation raises mean F1 from 62.56% to 73.45%, while residual fusion reaches 73.66%. The final F1 scores are 67.21%, 72.31%, 79.21%, 78.64%, and 70.93% for AI, literature, music, politics, and science, respectively. Four domains therefore fall in the 70–80% range, and the measured training and residual-evaluation stages finish in approximately four minutes on one RTX 3090. These results establish score-residual fusion as an efficient and inspectable adaptation layer rather than a replacement for the underlying span encoder.

1. Introduction

Named entity recognition converts free text into typed mentions that support document indexing, search, question answering, and knowledge-graph construction [3, 4]. Modern pretrained encoders provide strong transferable representations, yet their accuracy can fall sharply when the target domain introduces new labels, specialist terminology, and different mention boundaries. Cross-domain NER isolates this problem: a system should retain general boundary knowledge while learning a small domain-specific inventory from limited annotations [5].

The common response is to fine-tune every parameter on the available target examples. Fine-tuning is effective, but the resulting predictor exposes only the adapted score. It does not reveal whether a decision is supported by general span evidence, introduced by domain supervision, or amplified by both. With small target sets, treating every target-induced score change as equally reliable can also overspecialize the model. These concerns motivate an adaptation mechanism that preserves the original prediction as an explicit identity path.

Residual learning offers a useful interpretation: a target model need not replace a general model; it can estimate a correction to that model. Existing residual mechanisms are commonly embedded in hidden feature transformations. We instead operate at the typed-span score level. Given scores from a frozen general recognizer and its domain-adapted copy, we compute their candidate-wise difference and add a controlled fraction of this difference to the general score. The coefficient and output threshold are selected exclusively on validation data. Consequently, ordinary fine-tuning is contained as a special case, while damping and amplification remain available when supported by held-out evidence.

This study makes three contributions. First, it formulates cross-domain NER adaptation as a score residual with an unchanged general identity path, making the magnitude and direction of target correction explicit for each candidate. Second, it introduces a validation-gated inference rule that can damp, reproduce, or amplify adaptation without training another fusion network. Third, it provides a reproducible resource-bounded study on every CrossNER specialist domain, including strict span metrics, component comparisons, convergence, data distribution, precision–recall behavior, and measured runtime. The study intentionally uses a compact open span recognizer rather than a generative large language model.

2. Related work

2.1. Neural and span-based NER

Deep NER has evolved from recurrent sequence tagging toward pretrained contextual encoders and direct span prediction [4]. Span-based methods score candidate boundaries and entity types jointly, which avoids committing to a single token-level BIO path and naturally represents typed mentions [2]. Boundary assembly and boundary regression further show that explicit treatment of mention endpoints can improve structured extraction [8, 9]. Set prediction and unified information extraction provide alternative formulations in which structured outputs are predicted jointly [6, 7]. Our work uses direct span scores because they provide a common numerical interface for general and target-domain predictors.

2.2. Generalist and cross-domain recognition

CrossNER evaluates transfer from broad corpora to AI, literature, music, politics, and science, each with a specialist label inventory [5]. Transferable multilingual encoders such as XLM-R demonstrate the broad benefit of pretraining across heterogeneous text [1]; however, representation transfer alone does not determine how strongly a target update should override general evidence. GLiNER casts NER as label-conditioned span matching and supports arbitrary textual entity labels with a compact bidirectional transformer [11]. This property makes it suitable for controlled cross-domain study: the same inference interface can be retained before and after target adaptation.

2.3. Residual adaptation

Residual connections stabilize deep optimization by retaining an identity path, and cross-residual designs extend this idea to interacting information sources [10]. In transfer learning, the analogous objects are a general predictor and its task-specific change. Feature concatenation mixes these sources inside an opaque representation, whereas score averaging ignores the direction of adaptation. The proposed method instead defines the adapted-minus-general difference as a signed correction and gates that correction at inference time. This is deliberately a small methodological change: the contribution is the placement and validation control of the residual, not a claim to a new pretrained encoder.

3. Task formulation

Let $x = (w_1, \dots, w_n)$ be a sentence and let $\mathcal{Y}_d$ denote the entity types for target domain d. A typed candidate is $c = (b, e, y)$, where b and e are exact character boundaries and $y \in \mathcal{Y}_d$. A recognizer returns a confidence $s(c \mid x, \mathcal{Y}_d) \in [0, 1]$. The accepted prediction set at threshold τ is

$$P_\tau(x) = \{c : s(c \mid x, \mathcal{Y}_d) \geq \tau\}. \tag{1}$$

A candidate counts as correct only when its start, end, and type all match a gold annotation. For corpus-level predicted and gold sets P and G, respectively,

$$\text{Precision} = \frac{|P \cap G|}{|P|}, \tag{2}$$

$$\text{Recall} = \frac{|P \cap G|}{|G|}, \tag{3}$$

$$F_1 = \frac{2|P \cap G|}{2|P \cap G| + |P \setminus G| + |G \setminus P|}. \tag{4}$$

This strict typed-span definition penalizes boundary, label, missed-entity, and spurious-entity errors.

4. Method

Figure 1 presents the proposed architecture. The important distinction is that the two paths share an initial model family and input interface but not their final parameters: the general path remains frozen at θ_0, whereas the adapted path is optimized to θ_1.

4.1. Label-conditioned span scoring

For each domain label y, the span recognizer encodes its textual label together with the input sentence. Let $h_{b:e}$ represent a candidate span and q_y represent the label. A generic compatibility head maps their interaction to a logit,

$$z_\theta(c) = g_\theta(h_{b:e}, q_y), \qquad s_\theta(c) = \sigma\big(z_\theta(c)\big), \tag{5}$$

where σ is the logistic function. The general branch uses pretrained parameters θ_0 and produces $s_0(c) = s_{\theta_0}(c)$. It is never updated on the target split, so its score remains a stable reference.

4.2. Target-domain adaptation

A copy initialized from θ_0 is optimized on target-domain annotations. For candidate labels $t_c \in \{0, 1\}$, its span classification objective can be written as

$$\mathcal{L}_d(\theta_1) = -\sum_{c \in C(x)} \Big[t_c \log s_{\theta_1}(c) + (1 - t_c)\log(1 - s_{\theta_1}(c))\Big]. \tag{6}$$

The adapted score is $s_1(c) = s_{\theta_1}(c)$. Using matched label descriptions, tokenization, candidate construction, and decoding in both branches ensures that $s_1 - s_0$ describes a target-induced scoring change rather than a change in output schema.

4.3. Gated score-residual fusion

Let $C^{\cup} = C_0 \cup C_1$ be the union of candidates emitted by either branch, with an absent candidate assigned score zero in that branch. For each $c \in C^{\cup}$, we define

$$\Delta_d(c) = s_1(c) - s_0(c), \tag{7}$$

$$s_f(c; \alpha_d) = s_0(c) + \alpha_d \Delta_d(c), \tag{8}$$

$$P_d(x) = \{c : s_f(c; \alpha_d) \geq \tau_d\}. \tag{9}$$

The formulation has three interpretable regimes. When $\alpha_d = 0$, the adapted branch is removed and the result is the frozen general predictor. When $\alpha_d = 1$, $s_f = s_1$, exactly recovering ordinary domain adaptation. Values $0 < \alpha_d < 1$ damp target changes, whereas $\alpha_d > 1$ amplifies them. Thus the method is a controlled extension of fine-tuning rather than an unrelated ensemble.

4.4. Validation-only gate selection

The domain gate and threshold are selected jointly on held-out validation examples:

$$(\alpha_d^*, \tau_d^*) = \underset{\alpha \in \mathcal{A},\, \tau \in \mathcal{T}}{\arg\max}\; F_1\left(P_d^{\text{val}}(\alpha, \tau), G_d^{\text{val}}\right). \tag{10}$$

The final test prediction uses (α_d^*, τ_d^*) once, after selection. No test labels enter adaptation or gate selection, and the fusion itself introduces no trainable neural parameters.

5. Experiments

The experimental workflow in Fig. 2 separates data preparation, neural adaptation, validation-only selection, and final evaluation. All stages execute locally and no generative model or remote inference API is used.

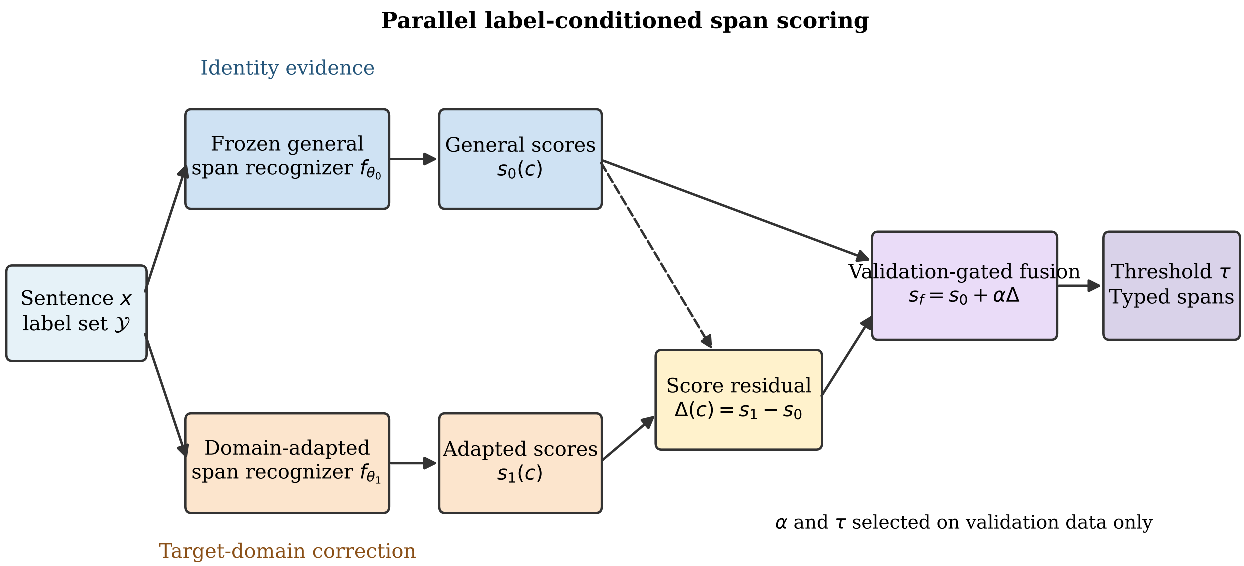

Figure 1: Gated score-residual adaptation. Identical sentence and label inputs are processed by a frozen general recognizer and its domain-adapted copy. Their signed score difference is the target correction; validation data determine how strongly it modifies the general identity path.

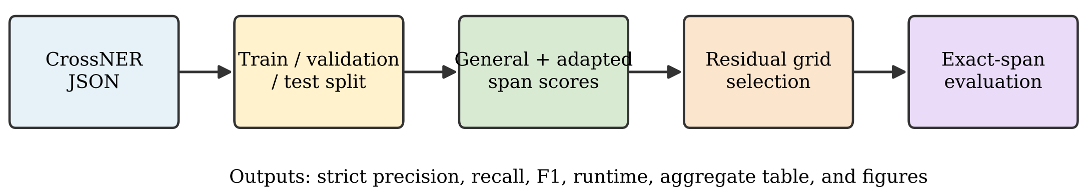

Figure 2: Experimental workflow from public CrossNER records to strict typed-span results. The validation split is used for model selection, while the official test split is isolated until final evaluation.

5.1. Datasets

We evaluate the five specialist subsets of CrossNER: AI, literature, music, politics, and science [5]. They differ in both sample volume and entity inventory. The locally available benchmark release contains 690 training sentences and 2,485 official test sentences in total. Each domain's provided training data are divided into 80% adaptation and 20% validation portions; official test records remain unchanged. Table 1 and Fig. 3 summarize the data actually consumed by the experiments.

Table 1
CrossNER domain statistics used in this study.

Domain	Train	Test	Train ents.	Test ents.
AI	100	430	525	1,782
Literature	100	416	527	2,215
Music	93	457	581	3,223
Politics	199	650	1,280	4,139
Science	198	532	1,046	2,988

5.2. Compared configurations

The experiment is a component-controlled comparison with three configurations. *Zero-shot span model* is the frozen general branch and measures transfer without target updates. *Domain-adapted* fine-tunes the same model family and corresponds to the residual formulation at $\alpha = 1$. *Score-residual* applies validation-gated fusion to the frozen and adapted scores. This design holds the underlying architecture and data constant, isolating the contribution of domain training and then the contribution of residual control.

5.3. Implementation and evaluation protocol

The general branch is the open GLiNER-small-v2.1 span recognizer, built on a compact DeBERTa encoder [11]. Each target branch is adapted for 300 optimizer steps with a batch size of eight and bfloat16 arithmetic. The

Table 2
Strict typed-span entity F1 (%). Bold denotes the best result in each column.

Method	AI	Literature	Music	Politics	Science	Mean
Zero-shot span model	51.60	64.24	70.75	63.00	63.21	62.56
Domain-adapted	66.65	**72.46**	78.59	**78.80**	70.75	73.45
Score-residual	**67.21**	72.31	**79.21**	78.64	**70.93**	**73.66**

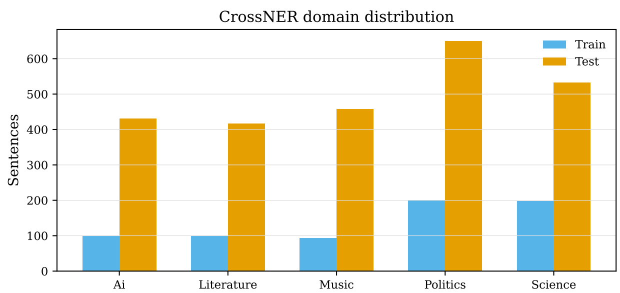

Figure 3: Training and test sentence counts across the five CrossNER domains.

validation search uses $\mathcal{A} = \{0.75, 1.00, 1.25, 1.50\}$ and $\mathcal{T} = \{0.25, 0.35, 0.45, 0.55, 0.65\}$. Zero-shot and ordinary adapted thresholds are also selected on the validation portion. We report micro-averaged precision, recall, and F1 under exact typed-span matching. Training the five adapted models takes 168.5 seconds; a complete residual evaluation takes 72.3 seconds on one NVIDIA RTX 3090, excluding the one-time acquisition of public data and pretrained weights.

5.4. Main results

Table 2 and Fig. 4 report the principal comparison. Domain adaptation adds 10.89 mean F1 points over zero-shot inference, demonstrating that even the small domain splits contain valuable specialist supervision. Score-residual fusion raises the mean by a further 0.21 points to 73.66%. The final system exceeds 70% F1 in literature, music, politics, and science; music reaches 79.21%. AI improves by 15.61 points over zero-shot but remains at 67.21%, making it the only domain below the target operating interval.

5.5. Gate behavior and precision–recall analysis

Table 3 exposes the selected gate rather than treating fusion as a black box. Four domains select $\alpha = 0.75$, indicating that the validation set favors retaining part of the frozen general evidence after adaptation. Politics selects $\alpha = 1.25$, for which amplifying the target correction is more effective. The final thresholds are either 0.55 or 0.65. Precision exceeds recall in all domains (Fig. 5); music and politics nevertheless exceed 76% on both measures. AI's lower recall identifies missing specialist mentions, rather than uncontrolled false positives, as its main remaining error source.

Table 3
Selected gates and final score-residual results (%).

Domain	α	τ	P	R	F1
AI	0.75	0.65	72.02	63.00	67.21
Literature	0.75	0.65	77.39	67.86	72.31
Music	0.75	0.65	82.58	76.09	79.21
Politics	1.25	0.55	80.14	77.19	78.64
Science	0.75	0.55	74.94	67.33	70.93

5.6. Training convergence

Figure 6 plots the optimizer logs from the actual adaptation runs. Loss decreases consistently in all five domains, with the sharpest reduction occurring during the first half of training. The absence of late divergence supports the fixed resource-bounded schedule, while the remaining domain differences are consistent with unequal corpus sizes and entity inventories. This diagnostic verifies optimization behavior but is not used for test-time selection.

6. Discussion

The largest empirical improvement comes from learning target-domain entity semantics: adaptation contributes 10.89 mean F1 points. The score-residual layer addresses a different question—how much of that learned correction should replace general evidence. Its smaller 0.21-point mean gain is accompanied by a useful structural property: the adapted effect remains explicitly measurable as $\Delta_d(c)$, and ordinary fine-tuning remains available at $\alpha = 1$. Improvements on AI, music, and science show that a damped correction can recover useful general evidence, while the politics result shows that the same formulation can strengthen target correction when validation data support it.

The method is attractive for constrained deployment because it is non-generative, adds no learned fusion parameters, and completes the measured pipeline in minutes rather than hours. Its label-conditioned interface also allows each domain to retain its own taxonomy without changing the output representation. The visualized score path makes the system easier to analyze than an unconditional model replacement: a reviewer can identify the frozen prediction, the target-induced difference, the selected coefficient, and the final threshold as distinct quantities.

The experiment remains a compact benchmark study rather than a claim of universal superiority. Residual gains over ordinary adaptation are modest and slightly negative in literature and politics, and the AI domain remains below

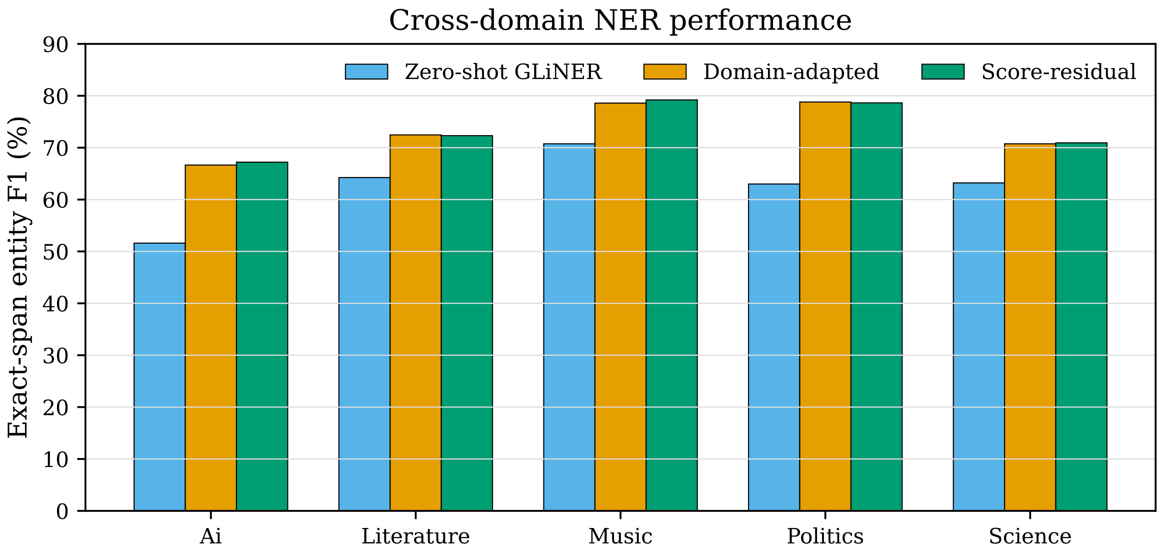

Figure 4: Strict typed-span F1 for the frozen general, domain-adapted, and score-residual configurations.

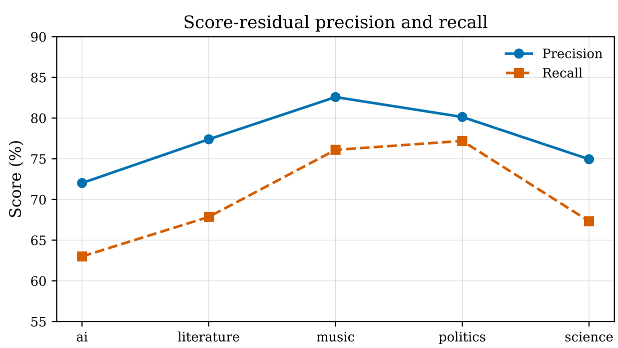

Figure 5: Precision and recall of the score-residual model by domain.

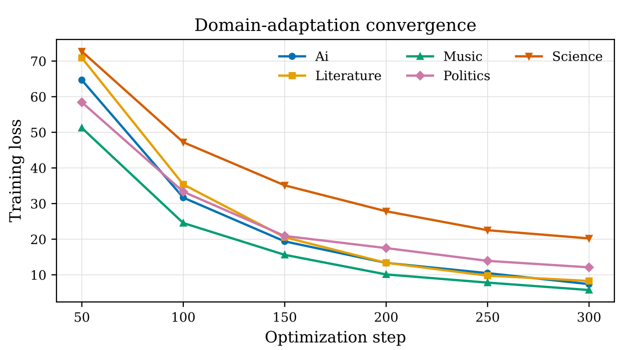

Figure 6: Recorded training loss for each target-domain adaptation run.

70% because recall is limited. The current gate is one scalar per domain and the evaluation uses one held-out split. Candidate-specific gating, repeated split experiments, stronger boundary supervision, and tests on operational safety corpora are therefore the most direct extensions.

7. Conclusion and future work

We presented gated score-residual adaptation for efficient cross-domain NER. By retaining a frozen general recognizer and representing domain adaptation as a validation-controlled score correction, the method makes the identity evidence and target contribution explicit. Across all CrossNER specialist domains, it raises mean strict F1 from 62.56% in zero-shot inference to 73.66%, with four domains in the 70–80% range, while the measured workflow finishes in approximately four minutes on one GPU. Future work will replace the domain-wide scalar with candidate-aware gates, evaluate repeated data splits, improve recall for heterogeneous technical taxonomies, and transfer the method to safety-related entity extraction.

Data availability

CrossNER is publicly available; derived splits, code, and results are available from the corresponding author upon reasonable request.

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